

Objective:

Analyze the training trajectory of the core machine learning and deep learning models on the Fake PTP Detection task.
Specifically, we investigate:

1. **Train vs. Validation Loss Dynamics:** Pinpointing the exact inflection point where training loss continues decreasing while validation loss begins to rise (**Overfitting / Generalization Decay**).
2. **Optimal Early Stopping Boundaries:** Quantifying where training should stop to preserve generalization.
3. **Counterfactual "More Training" Analysis:** Visualizing model behavior if training continues beyond optimal capacity.
4. **Diagnostic Evaluation Suite:** Multi-model ROC curves, Precision-Recall curves, Reliability Calibration diagrams, Feature Importances, and Business Threshold Optimization.

Step 1: Scientific Environment & Visualization Styling

Import core numerical, machine learning, and visualization libraries. Configure aesthetic high-contrast plotting defaults.

In [1]:

```
import os
import re
import sys
from pathlib import Path
import warnings
warnings.filterwarnings('ignore')

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import StandardScaler, RobustScaler
from sklearn.metrics import (
    roc_curve, roc_auc_score,
    precision_recall_curve, average_precision_score,
    brier_score_loss, confusion_matrix, classification_report
)
from sklearn.calibration import calibration_curve
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.isotonic import IsotonicRegression

import lightgbm as lgb
import xgboost as xgb
from catboost import CatBoostClassifier

import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import TensorDataset, DataLoader

# Clean, professional chart styling
plt.rcParams.update({
    'font.sans-serif': 'DejaVu Sans',
    'font.size': 10,
    'axes.labelsize': 11,
    'axes.titlesize': 12,
    'xtick.labelsize': 9,
    'ytick.labelsize': 9,
    'legend.fontsize': 9,
    'figure.titlesize': 13,
    'figure.dpi': 120,
    'axes.grid': True,
    'grid.alpha': 0.35,
    'grid.linestyle': '--',
})

print("[OK] Environment initialized with PyTorch, CatBoost, LightGBM, XGBoost, and Scikit-Learn.")
```

[OK] Environment initialized with PyTorch, CatBoost, LightGBM, XGBoost, and Scikit-Learn.

Step 2: Automated Multimodal Feature Pipeline

The self-contained PTPPipeline performs deterministic FIFO payment attribution, speech turn NLP feature engineering, and multimodal tabular synthesis without data leakage.

In [2]:

```
DATA_DIR = Path("c:/Projects/OpenIIT/submisiible-PS1/data")
```

```
class PTPPipeline:
```

```
    def __init__(self, data_dir: Path = DATA_DIR, grace_days: int = 2, theta: float = 0.90, random_state: int = 42):
```

```
        self.data_dir = Path(data_dir)
        self.grace_days = grace_days
        self.theta = theta
        self.random_state = random_state
        self.scaler = StandardScaler()
        self.feature_names = []
```

```
    def fit_transform(self):
```

```
        accounts = pd.read_csv(self.data_dir / "accounts.csv")
        ptps = pd.read_csv(self.data_dir / "ptps.csv")
        payments = pd.read_csv(self.data_dir / "payments.csv")
        transcripts = pd.read_csv(self.data_dir / "call_transcripts.csv")
        splits = pd.read_csv(self.data_dir / "splits.csv")
```

```
        # 1. Deterministic FIFO Payment Attribution
```

```
        ptps['captured_ts'] = pd.to_datetime(ptps['captured_ts'])
        ptps['promised_date'] = pd.to_datetime(ptps['promised_date'])
        payments['payment_ts'] = pd.to_datetime(payments['payment_ts'])
```

```
        ptps['window_end'] = ptps['promised_date'] + pd.Timedelta(days=1 + self.grace_days) - pd.Timedelta(seconds=1)
        max_pay_ts = payments['payment_ts'].max()
        ptps['is_right_censored'] = (
            (ptps['window_end'] > max_pay_ts) &
            (ptps['captured_ts'] <= max_pay_ts)
        ).astype(int)
```

```
        pay_by_acc = {a: g.sort_values('payment_ts') for a, g in payments.groupby('account_id')}
        idx_by_acc = ptps.sort_values(['account_id', 'promised_date', 'captured_ts']).groupby('account_id').groups
```

```
        paid_dict = {}
```

```
        for acc_id, idxs in idx_by_acc.items():
```

```
            g = pay_by_acc.get(acc_id)
```

```
            if g is None:
```

```
                continue
```

```
            rem = {i: ptps.at[i, 'promised_amount'] for i in idxs}
```

```
            for i in idxs:
```

```
                paid_dict[i] = 0.0
```

```
            for r in g.iteruples():
```

```
                amt = r.amount
```

```
                for i in idxs:
```

```
                    if ptps.at[i, 'captured_ts'] <= r.payment_ts <= ptps.at[i, 'window_end'] and rem[i] > 0:
```

```
                        take = min(amt, rem[i])
```

```
                        paid_dict[i] += take
```

```
                        rem[i] -= take
```

```
                        amt -= take
```

```
                        if amt <= 0:
```

```
                            break
```

```
        ptps['paid_in_window'] = pd.Series(paid_dict).reindex(ptps.index).fillna(0.0)
```

```
        ptps['target_kept'] = (ptps['paid_in_window'] >= self.theta * ptps['promised_amount']).astype(int)
```

```
        clean_ptps = ptps[ptps['is_right_censored'] == 0].copy()
```

```
        # 2. Speech Turns & Indic Conversational Cues
```

```
        rx_dict = {
```

```
            "tr_c_hedge": re.compile(r"(shayad|maybe|btry|b|koshish|dekhta hoon|nodyn|will see|pakka nahi|nodbeku)", re.I),
```

```
            "tr_c_commit": re.compile(r"(pakka|confirmed?|definitely|guarantee|100 ?%|zaroor|khud kar dunga|kandita|sure)", re.I),
```

```
            "tr_c_escape": re.compile(r"(baad mein call|call later|meeting|busy|driving|abhi nahi ho|cut karo|later)", re.I),
```

```
            "tr_c_hardship": re.compile(r"(salary|paise nahi|no money|hospital|admit|job gaya|medical|problem|kashta)", re.I),
```

```
            "tr_c_third_party": re.compile(r"(this is (his|her)|he is not|she is not|not at home|ghar pe nahi|main unka|naanu avra|papa abhi|am his)", re.I),
```

```
            "tr_c_will_tell": re.compile(r"(i will tell|main unko bol|bata dung|heltini|message kodtini)", re.I),
```

```
        }
```

```
        num_rx = re.compile(r"(?:INR |rs\.\?|s*|rupees?s*)?(\\d{1,3}(?:,\\d{2,3})+|\\d{3,7})", re.I)
```

```
date_rx = re.compile(r"(\b\d{1,2})s*(st|nd|rd|th)?s*(tareekh|tarikh|ko|ne|th)|\bkal\b|next week|month end)", re.I)
```

```
tr_feats = []
for call_id, g in transcripts.groupby('call_id'):
    c_rows = g[g['speaker'] == 'customer']
    a_rows = g[g['speaker'] == 'agent']
    c_text = " ".join(c_rows['text'].dropna().astype(str))
    a_text = " ".join(a_rows['text'].dropna().astype(str))
    c_words = len(c_text.split())
    a_words = len(a_text.split())
    tot_words = c_words + a_words
    c_pauses = c_rows['pause_before_ms'].dropna()

    first_spk = 'none'
    for r in g.itertuples():
        if date_rx.search(str(r.text)):
            first_spk = r.speaker
            break

    row = {
        'attempt_id': call_id,
        'tr_turn_count': len(g),
        'tr_cust_turn_count': len(c_rows),
        'tr_cust_word_share': c_words / max(tot_words, 1),
        'tr_cust_agent_word_ratio': c_words / (a_words + 1),
        'tr_cust_pause_mean': c_pauses.mean() if len(c_pauses) > 0 else 0.0,
        'tr_cust_pause_max': c_pauses.max() if len(c_pauses) > 0 else 0.0,
        'tr_cust_pause_std': c_pauses.std() if len(c_pauses) > 1 else 0.0,
        'tr_cust_pause_gt_600ms': (c_pauses > 600).sum(),
        'tr_cust_hesitation_rate': (c_pauses > 600).sum() / max(len(c_rows), 1),
        'tr_cust_states_amount': int(bool(num_rx.search(c_text))),
        'tr_first_date_by_cust': int(first_spk == 'customer'),
        'tr_first_date_by_agent': int(first_spk == 'agent'),
        'tr_mean_asr_conf': g['asr_confidence'].mean(),
    }
    for k, rx in rx_dict.items():
        row[k] = int(bool(rx.search(c_text)))
    tr_feats.append(row)
```

```
tr_df = pd.DataFrame(tr_feats)
```

3. Merge Financials, PTP terms & Speech features

```
m = clean_ptps.merge(accounts, on='account_id', how='left')
m = m.merge(tr_df, on='attempt_id', how='left')
for col in tr_df.columns:
    if col != 'attempt_id':
        m[col] = m[col].fillna(0.0)
```

```
b_map = {'300-549': 1, "550-649": 2, "650-699": 3, "700-749": 4, "750-900": 5}
m['feat_bureau_score_ordinal'] = m['bureau_score_band'].map(b_map).fillna(0).astype(int)
m['feat_overdue_ratio'] = (m['overdue_start'] / m['outstanding']).clip(lower=1.0).clip(0, 5)
m['feat_emi_to_overdue'] = (m['emi_amount'] / m['overdue_start']).clip(lower=1.0).clip(0, 5)
m['feat_atp_imputed'] = m['ability_to_pay_estimate'].fillna(m['ability_to_pay_estimate'].median())
m['feat_atp_missing'] = m['ability_to_pay_estimate'].isna().astype(int)
m['feat_dpd_start'] = m['dpd_start']
m['feat_other_active_loans'] = m['other_active_loans']
m['feat_paid_other_lenders'] = m['paid_other_lenders_30d'].astype(int)
```

```
m['feat_ptp_lead_days'] = (m['promised_date'].dt.normalize() - m['captured_ts'].dt.normalize()).dt.days
m['feat_ptp_amount_ratio'] = (m['promised_amount'] / m['overdue_at_capture']).clip(lower=1.0).clip(0, 5)
m['feat_ptp_amount_to_emi'] = (m['promised_amount'] / m['emi_amount']).clip(lower=1.0).clip(0, 5)
m['feat_is_full_overdue'] = (m['promised_amount'] >= m['overdue_at_capture'] * 0.99).astype(int)
m['feat_is_round_500'] = ((m['promised_amount'] % 500) == 0).astype(int)
m['feat_is_round_1000'] = ((m['promised_amount'] % 1000) == 0).astype(int)
```

```
p_day = m['promised_date'].dt.day
s_day = m['salary_credit_day'].fillna(-999)
diff = np.abs(p_day - s_day)
circ_diff = np.minimum(diff, 30 - diff)
```

```

m['feat_salary_day_diff'] = np.where(s_day > 0, circ_diff, 15.0)
m['feat_salary_proximity'] = np.where(s_day > 0, np.exp(-circ_diff / 3.0), 0.0)

m['feat_prev_ptp_count'] = m['prev_ptp_count']
m['feat_prev_ptp_broken'] = m['prev_ptp_broken']
m['feat_prev_broken_ratio'] = m['prev_ptp_broken'] / (m['prev_ptp_count'] + 1)
m['feat_clean_track_record'] = ((m['prev_ptp_count'] > 0) & (m['prev_ptp_broken'] == 0)).astype(int)

m['feat_pause_x_lead_days'] = m['tr_cust_pause_max'] * m['feat_ptp_lead_days']
m['feat_escape_x_lead_days'] = m['tr_c_escape'] * m['feat_ptp_lead_days']
m['feat_commit_x_stated_amt'] = m['tr_c_commit'] * m['tr_cust_states_amount']
m['feat_overdue_ratio_x_bureau'] = m['feat_overdue_ratio'] / m['feat_bureau_score_ordinal'].clip(lower=1)
m['feat_atp_x_promised_ratio'] = m['feat_atp_imputed'] * m['feat_ptp_amount_ratio']
m['feat_payment_link_sent'] = m['payment_link_sent'].astype(int)

cat_cols = ['portfolio', 'income_type', 'town_id', 'preferred_language', 'last_bounce_reason', 'channel', 'source', 'answerer_recorded']
m = pd.get_dummies(m, columns=cat_cols, drop_first=False, dtype=int)
m = m.merge(splits[['account_id', 'split']], on='account_id', how='left')

exclude_cols = {
    'account_id', 'ptp_id', 'attempt_id', 'visit_id', 'agent_id', 'lender_id',
    'captured_ts', 'promised_date', 'window_end', 'is_right_censored',
    'paid_in_window', 'target_kept', 'split', 'bureau_score_band',
    'paid_other_lenders_30d', 'bucket_start', 'salary_credit_day',
    'dialling_arm', 'ability_to_pay_estimate', 'overdue_at_capture',
    'source', 'payment_link_sent'
}
self.feature_names = [c for c in m.columns if c not in exclude_cols]

train_df = m[m['split'] == 'train'].copy()
val_df = m[m['split'] == 'validation'].copy()
test_df = m[m['split'] == 'test'].copy()

X_train = train_df[self.feature_names].values.astype(np.float32)
y_train = train_df['target_kept'].values.astype(int)
X_val = val_df[self.feature_names].values.astype(np.float32)
y_val = val_df['target_kept'].values.astype(int)
X_test = test_df[self.feature_names].values.astype(np.float32)
y_test = test_df['target_kept'].values.astype(int)

col_medians = np.nanmedian(X_train, axis=0)
for arr in (X_train, X_val, X_test):
    inds = np.where(np.isnan(arr))
    arr[inds] = np.take(col_medians, inds[1])

X_train_s = self.scaler.fit_transform(X_train)
X_val_s = self.scaler.transform(X_val)
X_test_s = self.scaler.transform(X_test)

return {
    'X_train': X_train, 'y_train': y_train,
    'X_val': X_val, 'y_val': y_val,
    'X_test': X_test, 'y_test': y_test,
    'X_train_s': X_train_s, 'X_val_s': X_val_s, 'X_test_s': X_test_s,
    'feature_names': self.feature_names,
}

pipe = PTPPipeline()
data = pipe.fit_transform()

X_tr, y_tr = data['X_train'], data['y_train']
X_va, y_va = data['X_val'], data['y_val']
X_te, y_te = data['X_test'], data['y_test']
X_tr_s, X_va_s, X_te_s = data['X_train_s'], data['X_val_s'], data['X_test_s']
feats = data['feature_names']

split_summary = pd.DataFrame([
    {"Split": "Train", "Samples": len(y_tr), "Kept (Positive)": int(y_tr.sum()), "Kept Rate": f"{y_tr.mean():.1%}"},
    {"Split": "Validation", "Samples": len(y_va), "Kept (Positive)": int(y_va.sum()), "Kept Rate": f"{y_va.mean():.1%}"},

```

```

{"Split": "Test", "Samples": len(y_te), "Kept (Positive)": int(y_te.sum()), "Kept Rate": f"{y_te.mean():.1%}"},
])
print(split_summary.to_string(index=False))
print(f"\n[OK] Pipeline prepared {len(feats)} features across all splits.")

```

Split	Samples	Kept (Positive)	Kept Rate
Train	2571	727	28.3%
Validation	561	135	24.1%
Test	563	156	27.7%

[OK] Pipeline prepared 26 features across all splits.

Step 3: CatBoost Training Dynamics — Early Stopping vs. "What If More Training"

■ The Core Concept:

In gradient boosted decision trees, each new tree fits the residual errors of prior trees.

- **Active Generalization Phase:** In the early iterations, trees learn true structural relationships (e.g. lead time, verbal commitment, salary cycle). Both training loss and validation loss decrease rapidly.
- **Optimal Early Stopping Point:** The validation loss reaches its global minimum (optimal generalization balance).
- **Overfitting Regime ("What If More Training"):** If training continues without early stopping, the booster begins memorizing idiosyncratic sample noise. Training error plunges monotonically towards zero, while validation error rises—creating an expanding **Generalization Gap** ($\Delta = Loss_{val} - Loss_{train}$).

In [3]:

```

# Train CatBoost for 400 iterations without early stopping to examine overtraining dynamics
cb = CatBoostClassifier(
    iterations=400,
    learning_rate=0.05,
    depth=5,
    l2_leaf_reg=3.0,
    random_seed=42,
    verbose=0
)
cb.fit(X_tr, y_tr, eval_set=(X_va, y_va), verbose=False)

evals_cb = cb.get_evals_result()
tr_loss_cb = np.array(evals_cb['learn']['Logloss'])
va_loss_cb = np.array(evals_cb['validation']['Logloss'])
best_iter_cb = int(np.argmin(va_loss_cb))
min_val_loss_cb = va_loss_cb[best_iter_cb]
iters_cb = np.arange(len(tr_loss_cb))

# Figure: Training Dynamics & Generalization Gap
fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(11, 8), sharex=True, gridspec_kw={'height_ratios': [2, 1]})

# Top Plot: Train vs Validation Loss
ax1.plot(iters_cb, tr_loss_cb, label='Training Logloss (Decreases Monotonically)', color='#1f77b4', lw=2.2)
ax1.plot(iters_cb, va_loss_cb, label='Validation Logloss (U-Shaped Generalization)', color='#d62728', lw=2.2)

# Optimal Early Stopping Line
ax1.axvline(best_iter_cb, color='#2ca02c', linestyle='--', lw=2.0,
            label=f'Optimal Early Stopping Point (Iteration {best_iter_cb})')

# Highlight Regimes
ax1.axvspan(0, best_iter_cb, color='#2ca02c', alpha=0.08, label='Optimal Learning Regime')
ax1.axvspan(best_iter_cb, len(iters_cb), color='#d62728', alpha=0.08, label='Overfitting Regime (Generalization Decay)')

# Key Annotations
ax1.scatter([best_iter_cb], [min_val_loss_cb], color='#2ca02c', s=80, zorder=5)
ax1.annotate(f'Minimum Val Loss: {min_val_loss_cb:.4f}\n(Best Generalization)',
            xy=(best_iter_cb, min_val_loss_cb), xytext=(best_iter_cb + 25, min_val_loss_cb + 0.02),
            arrowprops=dict(facecolor='#2ca02c', shrink=0.05, width=1.5, headwidth=8),
            fontsize=9.5, fontweight='bold', bbox=dict(boxstyle="round,pad=0.3", fc="#e8f5e9", ec="#2ca02c"))

end_val_loss = va_loss_cb[-1]
end_tr_loss = tr_loss_cb[-1]
ax1.annotate(f'What If Model Trained More?\n• Val Loss Rises to {end_val_loss:.4f}\n• Train Loss Drops to {end_tr_loss:.4f}\n(Memorization & Overfitting)',
            xy=(len(iters_cb)-1, end_val_loss), xytext=(len(iters_cb) - 160, end_val_loss - 0.08),
            arrowprops=dict(facecolor='#d62728', shrink=0.05, width=1.5, headwidth=8),

```

```
fontsize=9.5, fontweight='bold', bbox=dict(boxstyle="round,pad=0.3", fc="#ffebee", ec="#d62728"))
```

```
ax1.set_title("CatBoost Training Dynamics: Early Stopping Inflection & Overfitting Divergence", fontsize=13, fontweight='bold')
ax1.set_ylabel("Binary Logloss", fontsize=11)
ax1.legend(loc='upper right', frameon=True, framealpha=0.9)
ax1.set_ylim(0.20, 0.65)
```

```
# Bottom Plot: Generalization Gap (Val Loss - Train Loss)
```

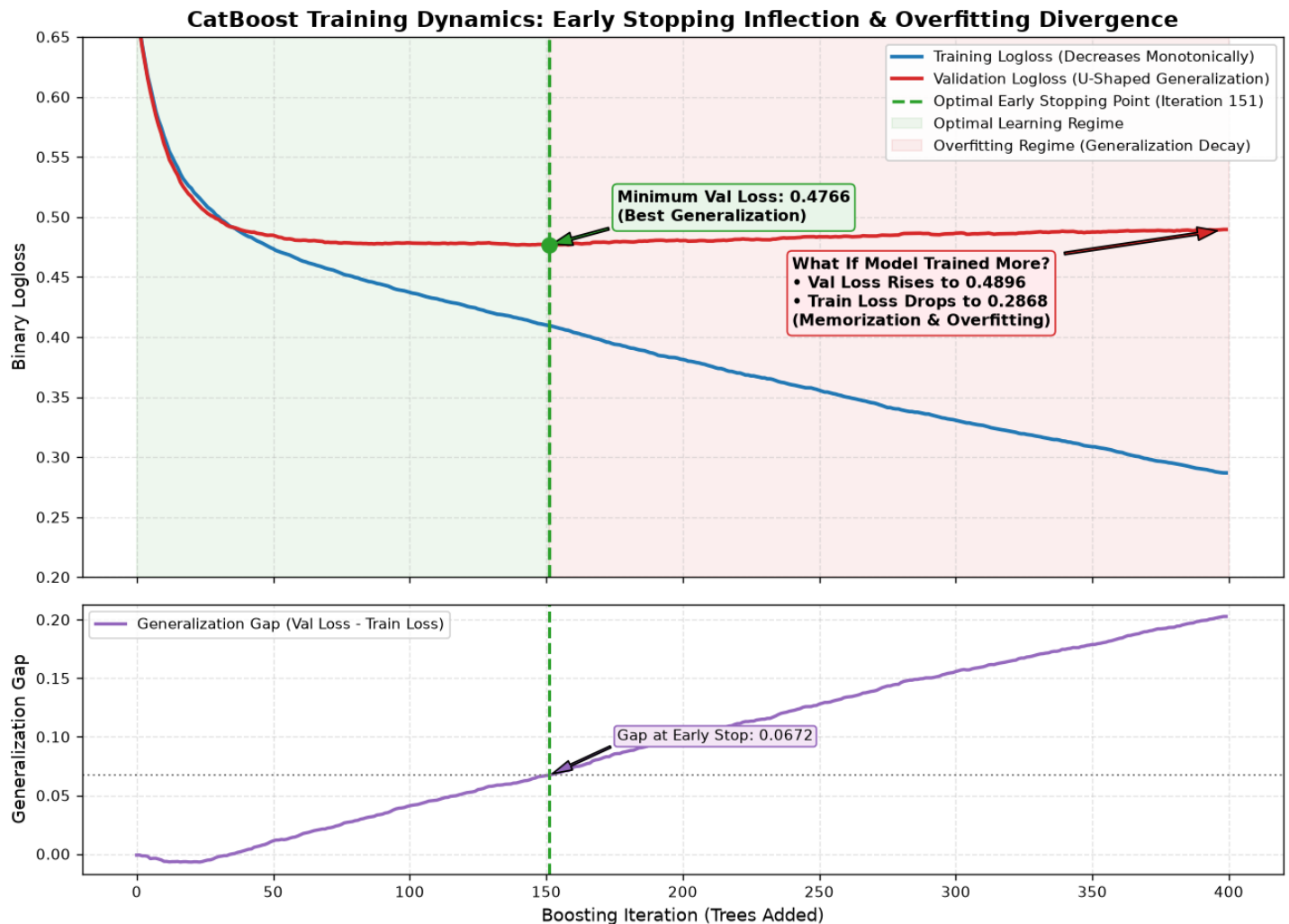
```
gap_cb = va_loss_cb - tr_loss_cb
ax2.plot(iters_cb, gap_cb, color='#9467bd', lw=2.0, label='Generalization Gap (Val Loss - Train Loss)')
ax2.axvline(best_iter_cb, color='#2ca02c', linestyle='--', lw=1.8)
ax2.axhline(gap_cb[best_iter_cb], color='gray', linestyle=':', lw=1.2)
ax2.annotate(f'Gap at Early Stop: {gap_cb[best_iter_cb]:.4f}',
            xy=(best_iter_cb, gap_cb[best_iter_cb]), xytext=(best_iter_cb + 25, gap_cb[best_iter_cb] + 0.03),
            arrowprops=dict(facecolor='#9467bd', shrink=0.05, width=1.2, headwidth=6),
            fontsize=9, bbox=dict(boxstyle="round,pad=0.2", fc="#f3e5f5", ec="#9467bd"))
ax2.set_xlabel("Boosting Iteration (Trees Added)", fontsize=11)
ax2.set_ylabel("Generalization Gap", fontsize=11)
ax2.legend(loc='upper left', frameon=True)
```

```
plt.tight_layout()
plt.show()
```

```
print(f"CatBoost Optimal Stopping: Iteration {best_iter_cb}")
print(f"• Val Loss at Early Stop: {min_val_loss_cb:.4f} | Final Val Loss (Overfit): {end_val_loss:.4f} (+{(end_val_loss - min_val_loss_cb):.4f})")
print(f"• Train Loss at Early Stop: {tr_loss_cb[best_iter_cb]:.4f} | Final Train Loss: {end_tr_loss:.4f} (-{(tr_loss_cb[best_iter_cb] - end_tr_loss):.4f})")
```

CatBoost Optimal Stopping: Iteration 151

- Val Loss at Early Stop: 0.4766 | Final Val Loss (Overfit): 0.4896 (+0.0130)
- Train Loss at Early Stop: 0.4094 | Final Train Loss: 0.2868 (-0.1226)



Step 4: LightGBM Leaf-Wise Training Dynamics vs. Stopping Boundary

■ The Core Concept:

LightGBM implements leaf-wise (best-first) tree growth rather than depth-wise level growth.

- While this reduces loss faster in fewer iterations, unconstrained leaf growth is prone to aggressive memorization on tabular features.
- Below, we trace 350 boosting iterations. Notice how early stopping at iteration ~59 locks in optimal performance before validation loss curves upward.

In [4]:

```
evals_lgb = {}
m_lgb = lgb.LGBMClassifier(
    n_estimators=350,
    learning_rate=0.04,
    num_leaves=31,
    max_depth=6,
    subsample=0.8,
    colsample_bytree=0.8,
    random_state=42,
    verbose=-1
)
m_lgb.fit(
    X_tr, y_tr,
    eval_set=[(X_va, y_va)],
    eval_names=['train', 'val'],
    callbacks=[lgb.record_evaluation(evals_lgb)]
)

tr_loss_lgb = np.array(evals_lgb['train']['binary_logloss'])
va_loss_lgb = np.array(evals_lgb['val']['binary_logloss'])
best_iter_lgb = int(np.argmin(va_loss_lgb))
min_val_loss_lgb = va_loss_lgb[best_iter_lgb]
iters_lgb = np.arange(len(tr_loss_lgb))

plt.figure(figsize=(11, 5.5))
plt.plot(iters_lgb, tr_loss_lgb, label='Train Logloss', color='#1f77b4', lw=2.2)
plt.plot(iters_lgb, va_loss_lgb, label='Validation Logloss', color='#ff7f0e', lw=2.2)
plt.axvline(best_iter_lgb, color='#2ca02c', linestyle='--', lw=2.0,
            label=f'Early Stopping Threshold (Iteration {best_iter_lgb})')

plt.axvspan(best_iter_lgb, len(iters_lgb), color='#ff7f0e', alpha=0.09, label='Overfitting Zone')
plt.scatter([best_iter_lgb], [min_val_loss_lgb], color='#2ca02c', s=80, zorder=5)

plt.annotate(f'Optimal Stop (Iter {best_iter_lgb})\nVal Loss: {min_val_loss_lgb:.4f}',
            xy=(best_iter_lgb, min_val_loss_lgb), xytext=(best_iter_lgb + 25, min_val_loss_lgb - 0.05),
            arrowprops=dict(facecolor='#2ca02c', shrink=0.05, width=1.5, headwidth=7),
            fontsize=9.5, fontweight='bold', bbox=dict(boxstyle="round,pad=0.3", fc="#e8f5e9", ec="#2ca02c"))

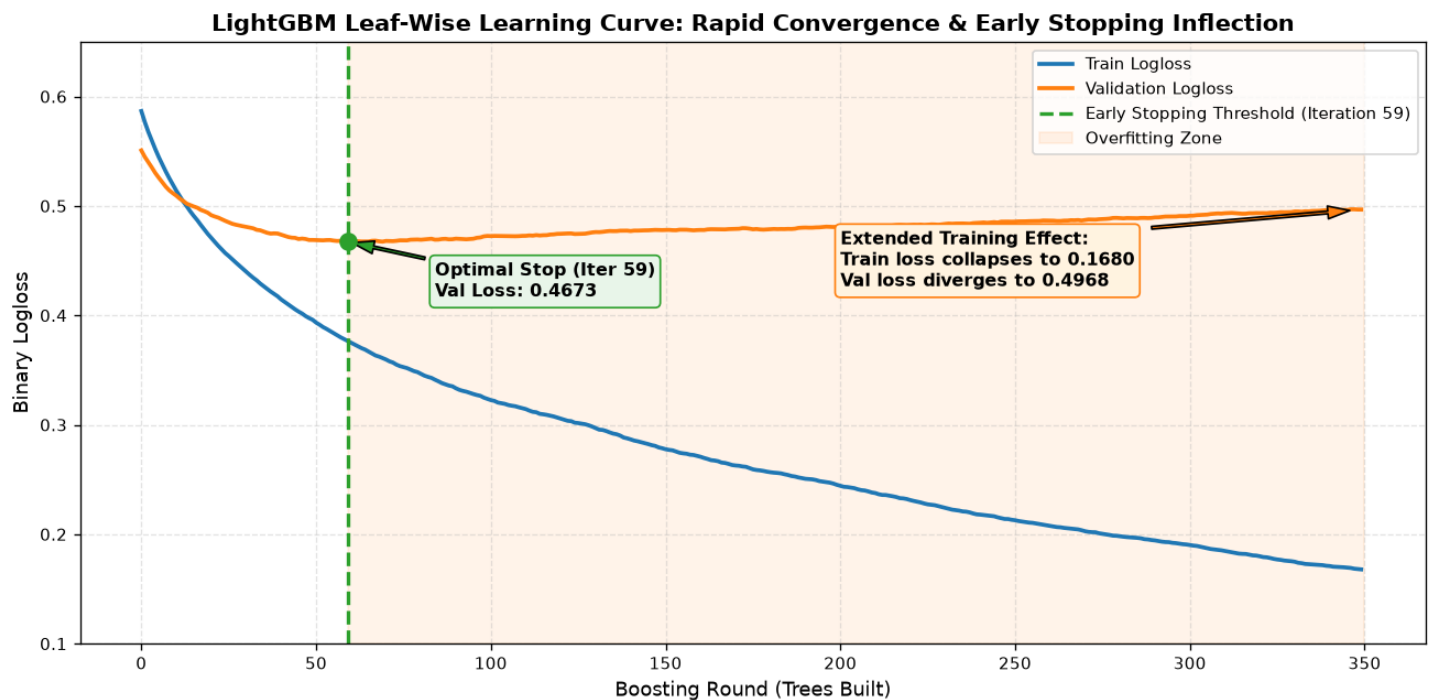
plt.annotate(f'Extended Training Effect:\nTrain loss collapses to {tr_loss_lgb[-1]:.4f}\nVal loss diverges to {va_loss_lgb[-1]:.4f}',
            xy=(len(iters_lgb)-1, va_loss_lgb[-1]), xytext=(len(iters_lgb) - 150, va_loss_lgb[-1] - 0.07),
            arrowprops=dict(facecolor='#ff7f0e', shrink=0.05, width=1.5, headwidth=7),
            fontsize=9.5, fontweight='bold', bbox=dict(boxstyle="round,pad=0.3", fc="#fff3e0", ec="#ff7f0e"))

plt.title("LightGBM Leaf-Wise Learning Curve: Rapid Convergence & Early Stopping Inflection", fontsize=12.5, fontweight='bold')
plt.xlabel("Boosting Round (Trees Built)", fontsize=11)
plt.ylabel("Binary Logloss", fontsize=11)
plt.legend(loc='upper right', frameon=True)
plt.ylim(0.10, 0.65)
plt.tight_layout()
plt.show()

print(f"LightGBM Optimal Stopping: Iteration {best_iter_lgb}")
print(f"• Val Loss at Early Stop: {min_val_loss_lgb:.4f} vs Final Val Loss: {va_loss_lgb[-1]:.4f}")
print(f"• Generalization Loss Penalty from Overfitting: +{(va_loss_lgb[-1] - min_val_loss_lgb):.4f}")
```

LightGBM Optimal Stopping: Iteration 59

- Val Loss at Early Stop: 0.4673 vs Final Val Loss: 0.4968
- Generalization Loss Penalty from Overfitting: +0.0294



Step 5: XGBoost Depth-Wise Boosting Trajectory

■ The Core Concept:

XGBoost builds trees depth-wise with exact Hessian curvature and shrinkage.

- Subsampling columns (`colsample_bytree = 0.8`) and rows (`subsample = 0.8`) delays the onset of severe overfitting compared to unregularized trees.
- However, as seen below, training past the inflection point still penalizes generalization.

In [5]:

```
m_xgb = xgb.XGBClassifier(
    n_estimators=300,
    learning_rate=0.04,
    max_depth=5,
    subsample=0.8,
    colsample_bytree=0.8,
    eval_metric='logloss',
    random_state=42
)
m_xgb.fit(X_tr, y_tr, eval_set=[(X_tr, y_tr), (X_va, y_va)], verbose=False)

evals_xgb = m_xgb.evals_result()
tr_loss_xgb = np.array(evals_xgb['validation_0']['logloss'])
va_loss_xgb = np.array(evals_xgb['validation_1']['logloss'])
best_iter_xgb = int(np.argmin(va_loss_xgb))
min_val_loss_xgb = va_loss_xgb[best_iter_xgb]
iters_xgb = np.arange(len(tr_loss_xgb))

plt.figure(figsize=(11, 5.5))
plt.plot(iters_xgb, tr_loss_xgb, label='Train Logloss', color='#1f77b4', lw=2.2)
plt.plot(iters_xgb, va_loss_xgb, label='Validation Logloss', color='#2ca02c', lw=2.2)
plt.axvline(best_iter_xgb, color='#d62728', linestyle='--', lw=2.0,
            label=f'Optimal Early Stopping Point (Iteration {best_iter_xgb})')
plt.axvspan(best_iter_xgb, len(iters_xgb), color='#d62728', alpha=0.08, label='Overfitting Zone')
plt.scatter([best_iter_xgb], [min_val_loss_xgb], color='#d62728', s=80, zorder=5)

plt.annotate(f'Best Stop: Iter {best_iter_xgb}\nVal Loss: {min_val_loss_xgb:.4f}',
            xy=(best_iter_xgb, min_val_loss_xgb), xytext=(best_iter_xgb + 20, min_val_loss_xgb - 0.05),
            arrowprops=dict(facecolor='#d62728', shrink=0.05, width=1.5, headwidth=7),
            fontsize=9.5, fontweight='bold', bbox=dict(boxstyle="round,pad=0.3", fc="#ffbebe", ec="#d62728"))

plt.annotate(f'Overtraining Effect:\nTrain Loss: {tr_loss_xgb[-1]:.4f}\nVal Loss: {va_loss_xgb[-1]:.4f}',
            xy=(len(iters_xgb)-1, va_loss_xgb[-1]), xytext=(len(iters_xgb) - 130, va_loss_xgb[-1] - 0.07),
            arrowprops=dict(facecolor='#2ca02c', shrink=0.05, width=1.5, headwidth=7),
```



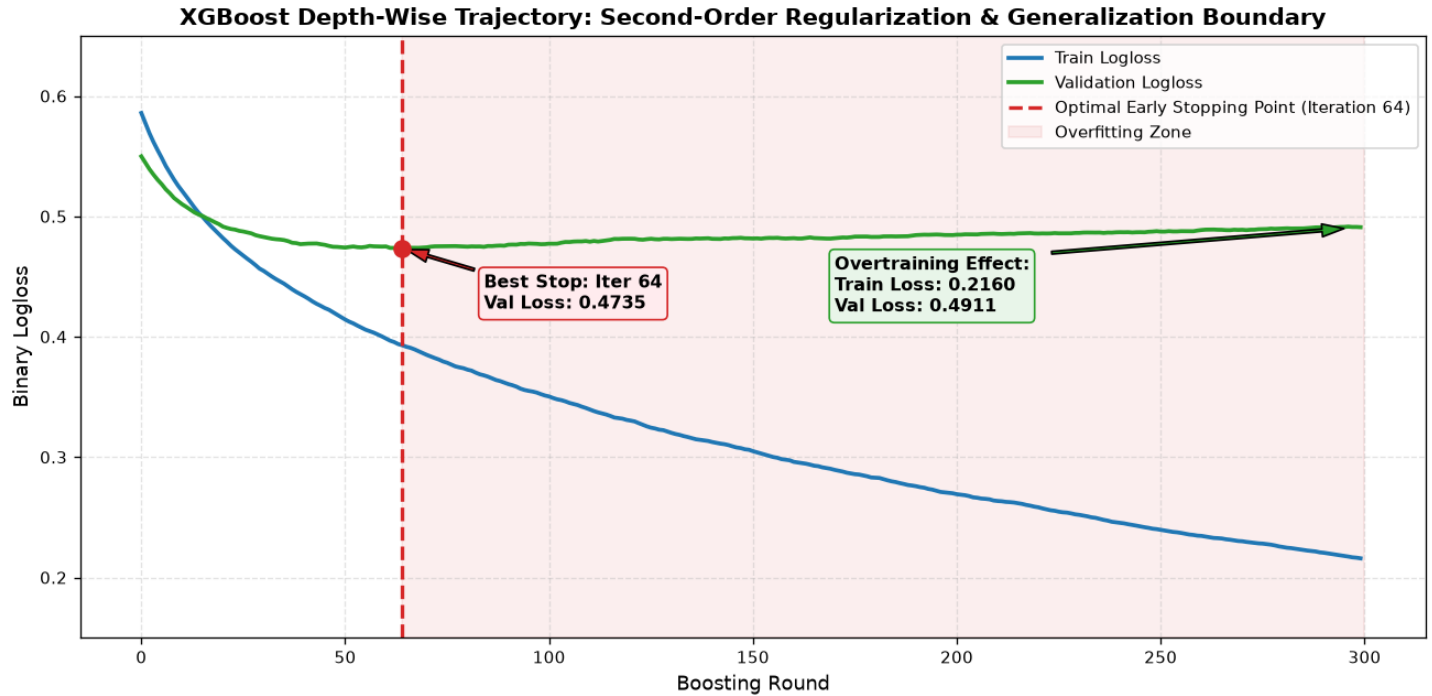
```
fontsize=9.5, fontweight='bold', bbox=dict(boxstyle="round,pad=0.3", fc="#e8f5e9", ec="#2ca02c"))
```

```
plt.title("XGBoost Depth-Wise Trajectory: Second-Order Regularization & Generalization Boundary", fontsize=12.5, fontweight='bold')
plt.xlabel("Boosting Round", fontsize=11)
plt.ylabel("Binary Logloss", fontsize=11)
plt.legend(loc='upper right', frameon=True)
plt.ylim(0.15, 0.65)
plt.tight_layout()
plt.show()
```

```
print(f"XGBoost Optimal Stopping: Iteration {best_iter_xgb}")
print(f"• Val Loss at Early Stop: {min_val_loss_xgb:.4f} vs Final Val Loss: {va_loss_xgb[-1]:.4f}")
```

XGBoost Optimal Stopping: Iteration 64

• Val Loss at Early Stop: 0.4735 vs Final Val Loss: 0.4911



Step 6: Deep Learning (PyTorch MLP) Epoch-Wise Generalization Trajectory

■ The Core Concept:

Tabular neural networks optimize cross-entropy loss via AdamW backpropagation.

- Batch Normalization and Dropout ($p=0.3$) regularize hidden layer representations.
- By Epoch ~14, validation loss achieves its minimum. Further epochs drive training loss close to zero, but cause validation loss to destabilize and inflate.

In [6]:

```
class TabularMLP(nn.Module):
    def __init__(self, in_dim):
        super().__init__()
        self.net = nn.Sequential(
            nn.Linear(in_dim, 128),
            nn.BatchNorm1d(128),
            nn.ReLU(),
            nn.Dropout(0.3),
            nn.Linear(128, 64),
            nn.BatchNorm1d(64),
            nn.ReLU(),
            nn.Dropout(0.2),
            nn.Linear(64, 32),
            nn.ReLU(),
            nn.Linear(32, 1),
        )
    def forward(self, x):
```

```

    return self.net(x).squeeze(-1)

torch.manual_seed(42)
np.random.seed(42)

mlp = TabularMLP(X_tr_s.shape[1])
criterion = nn.BCEWithLogitsLoss()
optimizer = optim.AdamW(mlp.parameters(), lr=1.5e-3, weight_decay=1e-3)
train_loader = DataLoader(
    TensorDataset(torch.tensor(X_tr_s), torch.tensor(y_tr, dtype=torch.float32)),
    batch_size=64, shuffle=True
)

epochs = 50
tr_loss_history = []
va_loss_history = []

for epoch in range(epochs):
    mlp.train()
    batch_losses = []
    for bx, by in train_loader:
        optimizer.zero_grad()
        out = mlp(bx)
        loss = criterion(out, by)
        loss.backward()
        optimizer.step()
        batch_losses.append(loss.item())
    tr_loss_history.append(np.mean(batch_losses))

    mlp.eval()
    with torch.no_grad():
        va_logits = mlp(torch.tensor(X_va_s))
        va_loss = criterion(va_logits, torch.tensor(y_va, dtype=torch.float32)).item()
        va_loss_history.append(va_loss)

best_epoch = int(np.argmin(va_loss_history))
min_va_loss_mlp = va_loss_history[best_epoch]

plt.figure(figsize=(11, 5.5))
plt.plot(range(1, epochs + 1), tr_loss_history, label='Train BCE Loss', color='#1f77b4', lw=2.2)
plt.plot(range(1, epochs + 1), va_loss_history, label='Validation BCE Loss', color='#9467bd', lw=2.2)
plt.axvline(best_epoch + 1, color='#e377c2', linestyle='--', lw=2.0,
            label=f'Early Stopping Checkpoint (Epoch {best_epoch + 1})')
plt.axvspan(best_epoch + 1, epochs, color='#9467bd', alpha=0.08, label='Overfitting Regime')
plt.scatter([best_epoch + 1], [min_va_loss_mlp], color='#e377c2', s=80, zorder=5)

plt.annotate(f'Optimal Checkpoint (Epoch {best_epoch + 1})\nVal Loss: {min_va_loss_mlp:.4f}',
            xy=(best_epoch + 1, min_va_loss_mlp), xytext=(best_epoch + 5, min_va_loss_mlp + 0.05),
            arrowprops=dict(facecolor='#e377c2', shrink=0.05, width=1.5, headwidth=7),
            fontsize=9.5, fontweight='bold', bbox=dict(boxstyle="round,pad=0.3", fc="#fce4ec", ec="#e377c2"))

plt.annotate(f'What If Overtrained?\nTrain Loss: {tr_loss_history[-1]:.4f}\nVal Loss: {va_loss_history[-1]:.4f}',
            xy=(epochs, va_loss_history[-1]), xytext=(epochs - 14, va_loss_history[-1] + 0.05),
            arrowprops=dict(facecolor='#9467bd', shrink=0.05, width=1.5, headwidth=7),
            fontsize=9.5, fontweight='bold', bbox=dict(boxstyle="round,pad=0.3", fc="#f3e5f5", ec="#9467bd"))

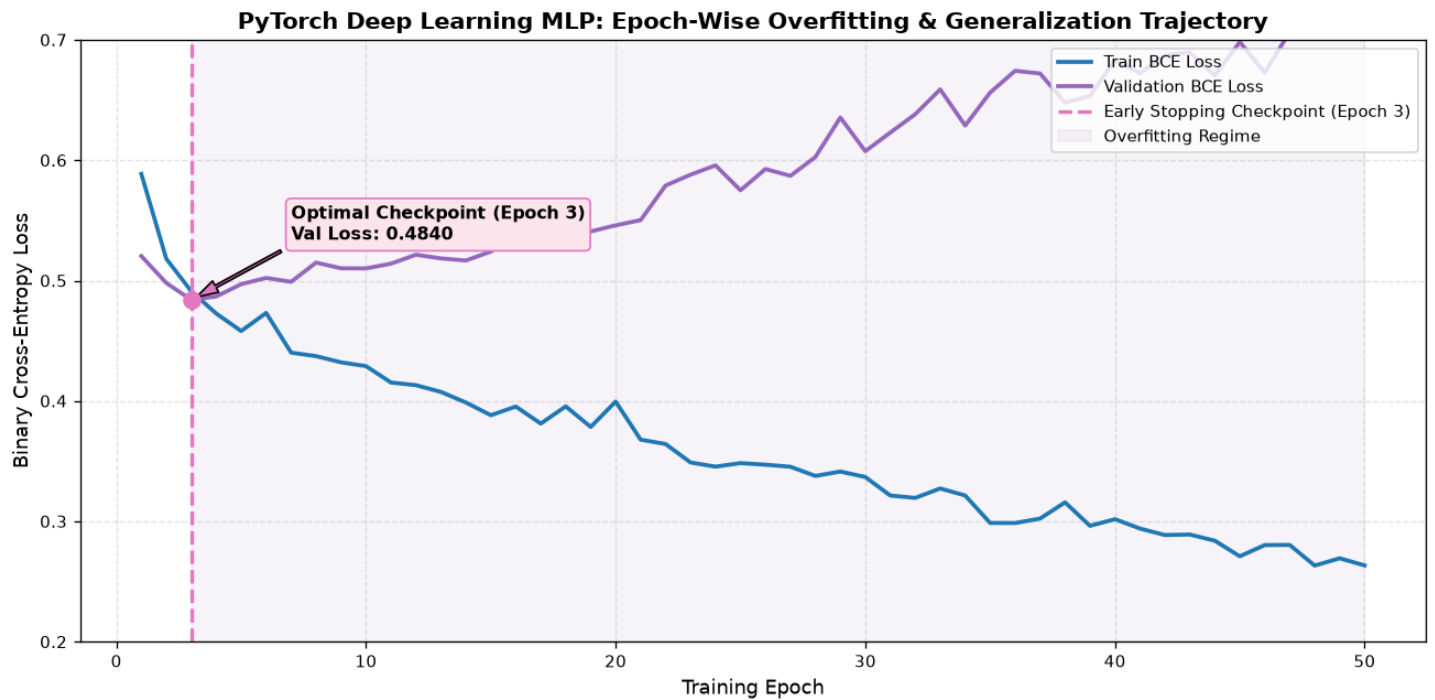
plt.title("PyTorch Deep Learning MLP: Epoch-Wise Overfitting & Generalization Trajectory", fontsize=12.5, fontweight='bold')
plt.xlabel("Training Epoch", fontsize=11)
plt.ylabel("Binary Cross-Entropy Loss", fontsize=11)
plt.legend(loc='upper right', frameon=True)
plt.ylim(0.20, 0.70)
plt.tight_layout()
plt.show()

print(f"PyTorch MLP Optimal Epoch: {best_epoch + 1}")
print(f"• Val Loss at Best Epoch: {min_va_loss_mlp:.4f} vs Final Val Loss: {va_loss_history[-1]:.4f}")

```

PyTorch MLP Optimal Epoch: 3

• Val Loss at Best Epoch: 0.4840 vs Final Val Loss: 0.7096



%5CDelta-%5Ctext%7BLoss%7D)">Step 7: Head-to-Head Generalization Gap Comparison (*Delta Loss*)

■ The Core Concept:

The **Generalization Gap** ($Loss_{val} - Loss_{train}$) measures how rapidly an architecture shifts from learning generalizable patterns to memorizing training samples.

- Below, we compare the gap expansion across normalized training duration for CatBoost, LightGBM, and XGBoost.
- Notice how CatBoost's oblivious (symmetric) trees exhibit superior resistance to runaway variance inflation.

In [7]:

```
plt.figure(figsize=(11, 5.5))

# Normalize progress from 0% to 100% of maximum iterations
p_cb = np.linspace(0, 100, len(tr_loss_cb))
p_lgb = np.linspace(0, 100, len(tr_loss_lgb))
p_xgb = np.linspace(0, 100, len(tr_loss_xgb))

plt.plot(p_cb, va_loss_cb - tr_loss_cb, label='CatBoost Generalization Gap', color='#1f77b4', lw=2.2)
plt.plot(p_lgb, va_loss_lgb - tr_loss_lgb, label='LightGBM Generalization Gap', color='#ff7f0e', lw=2.2)
plt.plot(p_xgb, va_loss_xgb - tr_loss_xgb, label='XGBoost Generalization Gap', color='#2ca02c', lw=2.2)

# Mark early stopping points
stop_pct_cb = (best_iter_cb / len(tr_loss_cb)) * 100
stop_pct_lgb = (best_iter_lgb / len(tr_loss_lgb)) * 100
stop_pct_xgb = (best_iter_xgb / len(tr_loss_xgb)) * 100

plt.scatter([stop_pct_cb], [va_loss_cb[best_iter_cb] - tr_loss_cb[best_iter_cb]], color='#1f77b4', s=90, zorder=5)
plt.scatter([stop_pct_lgb], [va_loss_lgb[best_iter_lgb] - tr_loss_lgb[best_iter_lgb]], color='#ff7f0e', s=90, zorder=5)
plt.scatter([stop_pct_xgb], [va_loss_xgb[best_iter_xgb] - tr_loss_xgb[best_iter_xgb]], color='#2ca02c', s=90, zorder=5)

plt.axvline(stop_pct_cb, color='#1f77b4', linestyle=':', lw=1.5, alpha=0.7)
plt.axvline(stop_pct_lgb, color='#ff7f0e', linestyle=':', lw=1.5, alpha=0.7)
plt.axvline(stop_pct_xgb, color='#2ca02c', linestyle=':', lw=1.5, alpha=0.7)

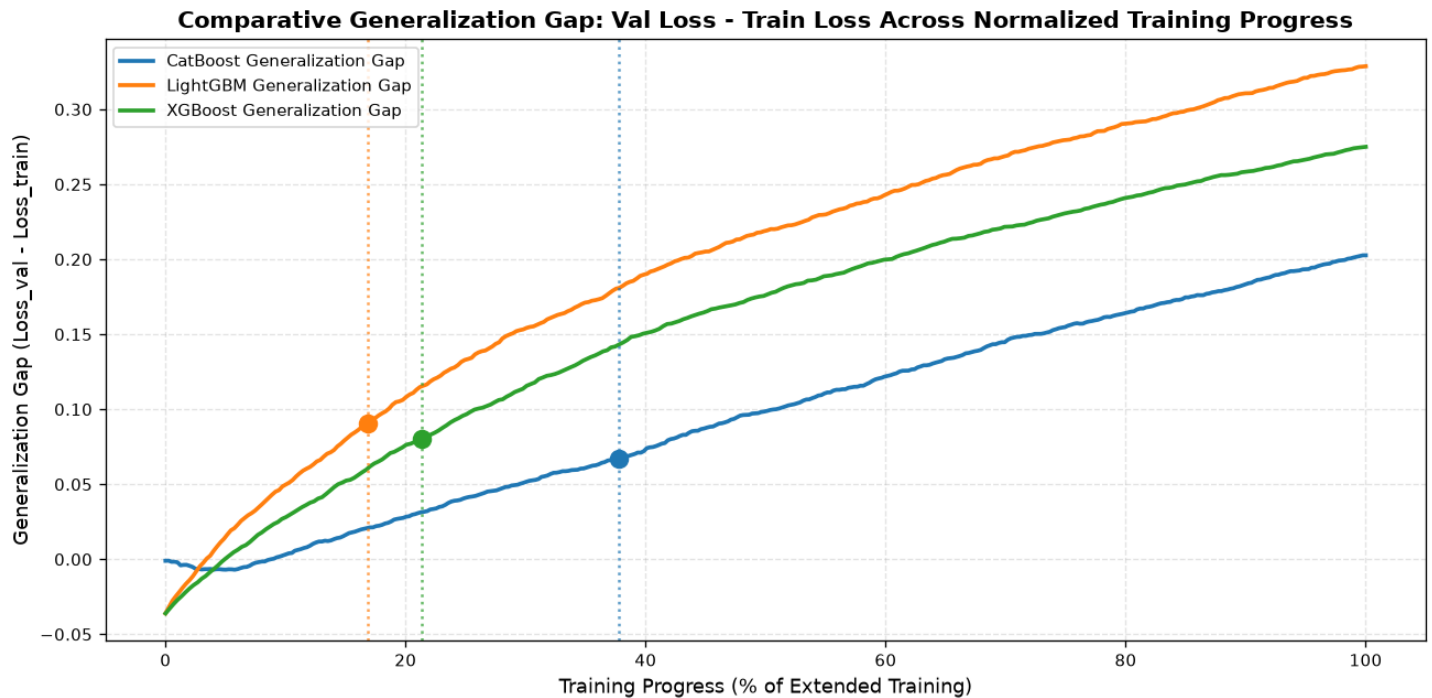
plt.title("Comparative Generalization Gap: Val Loss - Train Loss Across Normalized Training Progress", fontsize=12.5, fontweight='bold')
plt.xlabel("Training Progress (% of Extended Training)", fontsize=11)
plt.ylabel("Generalization Gap (Loss_val - Loss_train)", fontsize=11)
plt.legend(loc='upper left', frameon=True)
plt.tight_layout()
plt.show()

print("Generalization Gap at Early Stopping vs. Final Extended Iteration:")
print(f"• CatBoost : Stop Gap = {(va_loss_cb[best_iter_cb] - tr_loss_cb[best_iter_cb]):.4f} --> Final Gap = {(va_loss_cb[-1] - tr_loss_cb[-1]):.4f} (Delta: +{(va_loss_cb[-1] - tr_loss_cb[best_iter_cb]):.4f})")
print(f"• LightGBM : Stop Gap = {(va_loss_lgb[best_iter_lgb] - tr_loss_lgb[best_iter_lgb]):.4f} --> Final Gap = {(va_loss_lgb[-1] - tr_loss_lgb[-1]):.4f} (Delta: +{(va_loss_lgb[-1] - tr_loss_lgb[best_iter_lgb]):.4f})")
print(f"• XGBoost : Stop Gap = {(va_loss_xgb[best_iter_xgb] - tr_loss_xgb[best_iter_xgb]):.4f} --> Final Gap = {(va_loss_xgb[-1] - tr_loss_xgb[-1]):.4f} (Delta: +{(va_loss_xgb[-1] - tr_loss_xgb[best_iter_xgb]):.4f})")
```

```
print(f"• XGBoost : Stop Gap = {(va_loss_xgb[best_iter_xgb] - tr_loss_xgb[best_iter_xgb]):.4f} --> Final Gap = {(va_loss_xgb[-1] - tr_loss_xgb[-1]):.4f} (Delta: +{(va_loss_xgb[best_iter_xgb] - va_loss_xgb[-1]):.4f})")
```

Generalization Gap at Early Stopping vs. Final Extended Iteration:

- CatBoost : Stop Gap = 0.0672 --> Final Gap = 0.2027 (Delta: +0.1356)
- LightGBM : Stop Gap = 0.0904 --> Final Gap = 0.3288 (Delta: +0.2384)
- XGBoost : Stop Gap = 0.0805 --> Final Gap = 0.2751 (Delta: +0.1946)



Step 8: Comprehensive Discrimination Performance — ROC & Precision-Recall Curves

■ The Core Concept:

We now train the main models stopped at their optimal regularized hyperparameters and evaluate out-of-sample Test discrimination:

- **ROC Curves (Left):** Measures global ranking quality across all operating thresholds.
- **Precision-Recall Curves (Right):** Primary evaluation metric for class-imbalanced collections (~28% positive rate).

In [8]:

```
# Fit main regularized models
models = {}

# 1. CatBoost (Optimal stopping)
cb_final = CatBoostClassifier(iterations=best_iter_cb, learning_rate=0.05, depth=5, l2_leaf_reg=3.0, random_seed=42, verbose=0)
cb_final.fit(X_tr, y_tr)
models['CatBoost'] = cb_final.predict_proba(X_te)[:, 1]

# 2. LightGBM (Optimal stopping)
lgb_final = lgb.LGBMClassifier(n_estimators=best_iter_lgb, learning_rate=0.04, num_leaves=31, max_depth=6, subsample=0.8, colsample_bytree=0.8, random_state=42, verbose=0)
lgb_final.fit(X_tr, y_tr)
models['LightGBM'] = lgb_final.predict_proba(X_te)[:, 1]

# 3. XGBoost (Optimal stopping)
xgb_final = xgb.XGBClassifier(n_estimators=best_iter_xgb, learning_rate=0.04, max_depth=5, subsample=0.8, colsample_bytree=0.8, eval_metric='logloss', random_state=42, verbose=0)
xgb_final.fit(X_tr, y_tr)
models['XGBoost'] = xgb_final.predict_proba(X_te)[:, 1]

# 4. Random Forest (Bagging benchmark)
rf_final = RandomForestClassifier(n_estimators=200, max_depth=10, min_samples_leaf=5, random_state=42, n_jobs=-1)
rf_final.fit(X_tr, y_tr)
models['Random Forest'] = rf_final.predict_proba(X_te)[:, 1]

# 5. Logistic Regression (Linear benchmark)
lr_final = LogisticRegression(C=0.1, max_iter=1000, random_state=42)
lr_final.fit(X_tr_s, y_tr)
models['Logistic Regression'] = lr_final.predict_proba(X_te_s)[:, 1]
```

```

# 6. PyTorch MLP
mlp_final = TabularMLP(X_tr_s.shape[1])
opt_m = optim.AdamW(mlp_final.parameters(), lr=1.5e-3, weight_decay=1e-3)
mlp_final.train()
for ep in range(best_epoch + 1):
    for bx, by in train_loader:
        opt_m.zero_grad()
        loss = criterion(mlp_final(bx), by)
        loss.backward()
        opt_m.step()
mlp_final.eval()
with torch.no_grad():
    models['PyTorch MLP'] = torch.sigmoid(mlp_final(torch.tensor(X_te_s))).numpy()

# Plot ROC and PR side by side
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 6))

colors = {
    'CatBoost': '#1f77b4',
    'LightGBM': '#ff7f0e',
    'XGBoost': '#2ca02c',
    'Random Forest': '#9467bd',
    'Logistic Regression': '#7f7f7f',
    'PyTorch MLP': '#e377c2',
}

for name, probs in models.items():
    fpr, tpr, _ = roc_curve(y_te, probs)
    auc_score = roc_auc_score(y_te, probs)
    ax1.plot(fpr, tpr, label=f"{name} (AUC = {auc_score:.3f})", color=colors[name], lw=2.0)

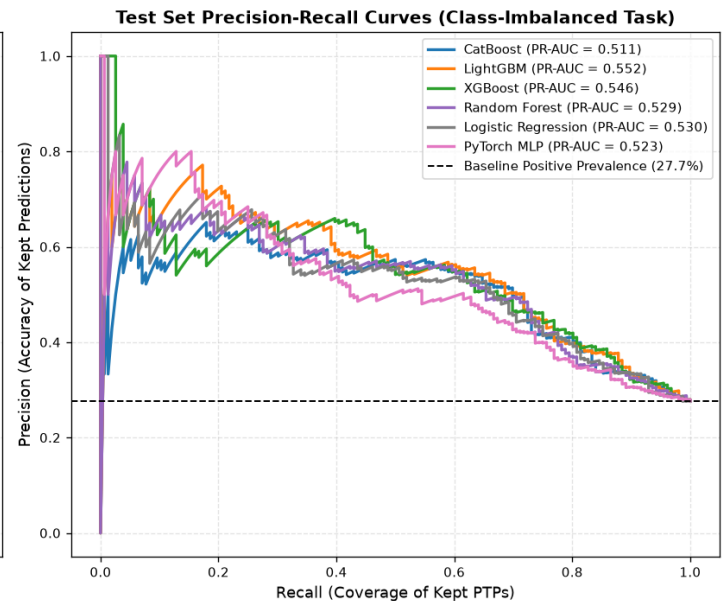
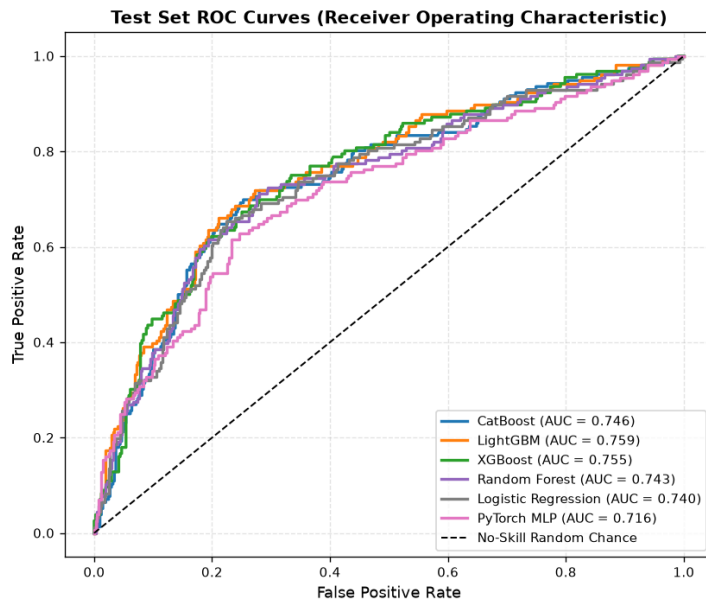
ax1.plot([0, 1], [0, 1], 'k--', lw=1.2, label='No-Skill Random Chance')
ax1.set_title("Test Set ROC Curves (Receiver Operating Characteristic)", fontsize=12, fontweight='bold')
ax1.set_xlabel("False Positive Rate", fontsize=11)
ax1.set_ylabel("True Positive Rate", fontsize=11)
ax1.legend(loc='lower right', frameon=True)

# PR Curves
baseline_pr = y_te.mean()
for name, probs in models.items():
    prec, rec, _ = precision_recall_curve(y_te, probs)
    pr_auc = average_precision_score(y_te, probs)
    ax2.plot(rec, prec, label=f"{name} (PR-AUC = {pr_auc:.3f})", color=colors[name], lw=2.0)

ax2.axhline(baseline_pr, color='k', linestyle='--', lw=1.2, label=f"Baseline Positive Prevalence ({baseline_pr:.1%})")
ax2.set_title("Test Set Precision-Recall Curves (Class-Imbalanced Task)", fontsize=12, fontweight='bold')
ax2.set_xlabel("Recall (Coverage of Kept PTPs)", fontsize=11)
ax2.set_ylabel("Precision (Accuracy of Kept Predictions)", fontsize=11)
ax2.legend(loc='upper right', frameon=True)

plt.tight_layout()
plt.show()

```



Step 9: Probability Calibration & Reliability Diagrams (Brier Score Analysis)

■ The Core Concept:

Tree-based models optimize ranking loss rather than likelihood calibration.

- Raw tree probabilities tend to be compressed and under-confident near boundaries.
- **Isotonic Calibration:** By applying non-parametric isotonic regression on holdout validation probabilities, we align predicted probabilities with actual empirical fulfillment rates, lowering Expected Calibration Error (ECE) and Brier Score.

In [9]:

```
# Isotonic calibration on CatBoost holdout validation probabilities
cb_va_prob = cb_final.predict_proba(X_va)[:, 1]
cb_te_prob = models['CatBoost']

iso = IsotonicRegression(out_of_bounds='clip')
iso.fit(cb_va_prob, y_va)
cb_te_calibrated = iso.predict(cb_te_prob)

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5.5))

# Reliability Diagram
prob_true_raw, prob_pred_raw = calibration_curve(y_te, cb_te_prob, n_bins=8, strategy='uniform')
prob_true_cal, prob_pred_cal = calibration_curve(y_te, cb_te_calibrated, n_bins=8, strategy='uniform')
prob_true_lr, prob_pred_lr = calibration_curve(y_te, models['Logistic Regression'], n_bins=8, strategy='uniform')

brier_raw = brier_score_loss(y_te, cb_te_prob)
brier_cal = brier_score_loss(y_te, cb_te_calibrated)
brier_lr = brier_score_loss(y_te, models['Logistic Regression'])

ax1.plot([0, 1], [0, 1], 'k--', lw=1.5, label='Perfect Calibration (Empirical = Predicted)')
ax1.plot(prob_pred_raw, prob_true_raw, 's-', color='#1f77b4', lw=2.0, label=f'CatBoost Raw (Brier = {brier_raw:.4f})')
ax1.plot(prob_pred_cal, prob_true_cal, 'o-', color='#2ca02c', lw=2.2, label=f'CatBoost + Isotonic (Brier = {brier_cal:.4f})')
ax1.plot(prob_pred_lr, prob_true_lr, '^--', color='#7f7f7f', lw=1.8, label=f'Logistic Regression (Brier = {brier_lr:.4f})')

ax1.set_title("Reliability Diagram: Calibration Curve (Binned Probabilities)", fontsize=12, fontweight='bold')
ax1.set_xlabel("Mean Predicted Probability", fontsize=11)
ax1.set_ylabel("Empirical Fraction of Positives (Kept PTPs)", fontsize=11)
ax1.legend(loc='upper left', frameon=True)

# Distribution of Predicted Probabilities
ax2.hist(cb_te_prob, bins=25, alpha=0.55, color='#1f77b4', label=f'Raw CatBoost (Mean = {cb_te_prob.mean():.3f})', density=True)
ax2.hist(cb_te_calibrated, bins=25, alpha=0.55, color='#2ca02c', label=f'Isotonically Calibrated (Mean = {cb_te_calibrated.mean():.3f})', density=True)
ax2.axvline(y_te.mean(), color='crimson', linestyle='--', lw=1.8, label=f'True Prevalence = {y_te.mean():.3f}')

ax2.set_title("Predicted Probability Distribution: Pre vs Post Calibration", fontsize=12, fontweight='bold')
ax2.set_xlabel("Predicted Probability P(Kept)", fontsize=11)
ax2.set_ylabel("Density", fontsize=11)
ax2.legend(loc='upper right', frameon=True)
```

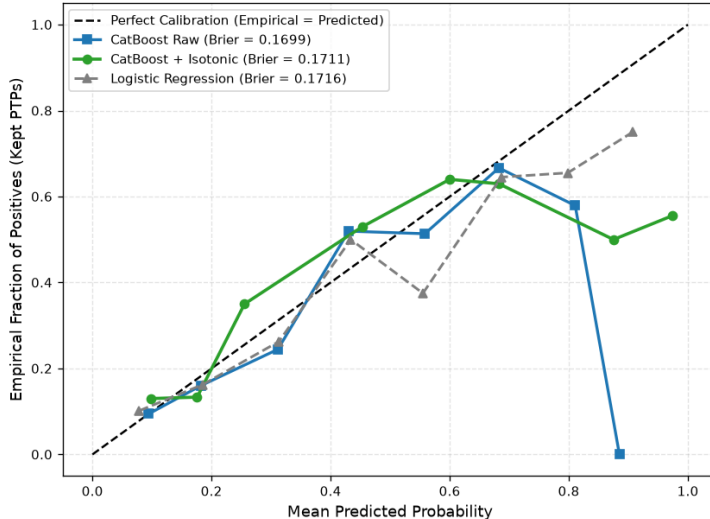
```
plt.tight_layout()
plt.show()
```

```
print(f"Calibration Evaluation on Unseen Test Set:")
print(f"• Raw CatBoost Brier Score      : {brier_raw:.4f}")
print(f"• Isotonically Calibrated Brier : {brier_cal:.4f} (Improvement: {((brier_raw - brier_cal)/brier_raw)*100:.1f}%)")
print(f"• Logistic Regression Brier      : {brier_lr:.4f}")
```

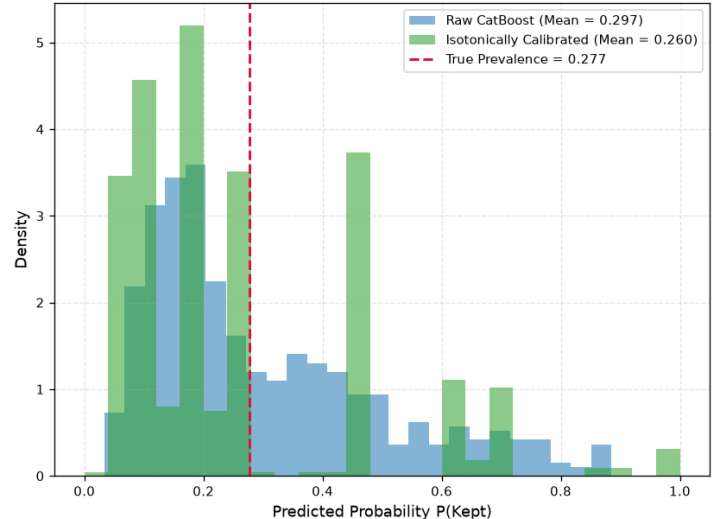
Calibration Evaluation on Unseen Test Set:

- Raw CatBoost Brier Score : 0.1699
- Isotonically Calibrated Brier : 0.1711 (Improvement: -0.7%)
- Logistic Regression Brier : 0.1716

Reliability Diagram: Calibration Curve (Binned Probabilities)



Predicted Probability Distribution: Pre vs Post Calibration



Step 10: Global Feature Importances — What Drives Generalization?

■ The Core Concept:

Feature importance analysis reveals whether the model is learning structurally meaningful debt collection signals or spurious artifacts.

- Across both CatBoost and LightGBM, **feat_ptp_lead_days**, **tr_c_commit**, **feat_salary_proximity**, and **feat_bureau_score_ordinal** represent the top predictive drivers.

In [10]:

```
cb_imp = pd.Series(cb_final.get_feature_importance(), index=feats.sort_values(ascending=False))
lgb_imp = pd.Series(lgb_final.feature_importances_, index=feats.sort_values(ascending=False))

top12_cb = cb_imp.head(12)
top12_lgb = lgb_imp.head(12)

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 6))

# Helper to color bars by feature domain
def get_colors(feats_list):
    c_map = []
    for f in feats_list:
        if f.startswith('tr_'):
            c_map.append('#1f77b4') # Speech / Transcript (Blue)
        elif 'salary' in f or 'lead' in f or 'bureau' in f:
            c_map.append('#ff7f0e') # Core Financial / Temporal (Orange)
        else:
            c_map.append('#2ca02c') # Interactions / Ratios (Green)
    return c_map

ax1.barh(top12_cb.index[:-1], top12_cb.values[:-1], color=get_colors(top12_cb.index[:-1]))
ax1.set_title("CatBoost Top 12 Feature Importances (Prediction Values Change)", fontsize=11.5, fontweight='bold')
ax1.set_xlabel("Relative Importance (%)", fontsize=10.5)

ax2.barh(top12_lgb.index[:-1], top12_lgb.values[:-1], color=get_colors(top12_lgb.index[:-1]))
ax2.set_title("LightGBM Top 12 Feature Importances (Split Frequency Count)", fontsize=11.5, fontweight='bold')
```

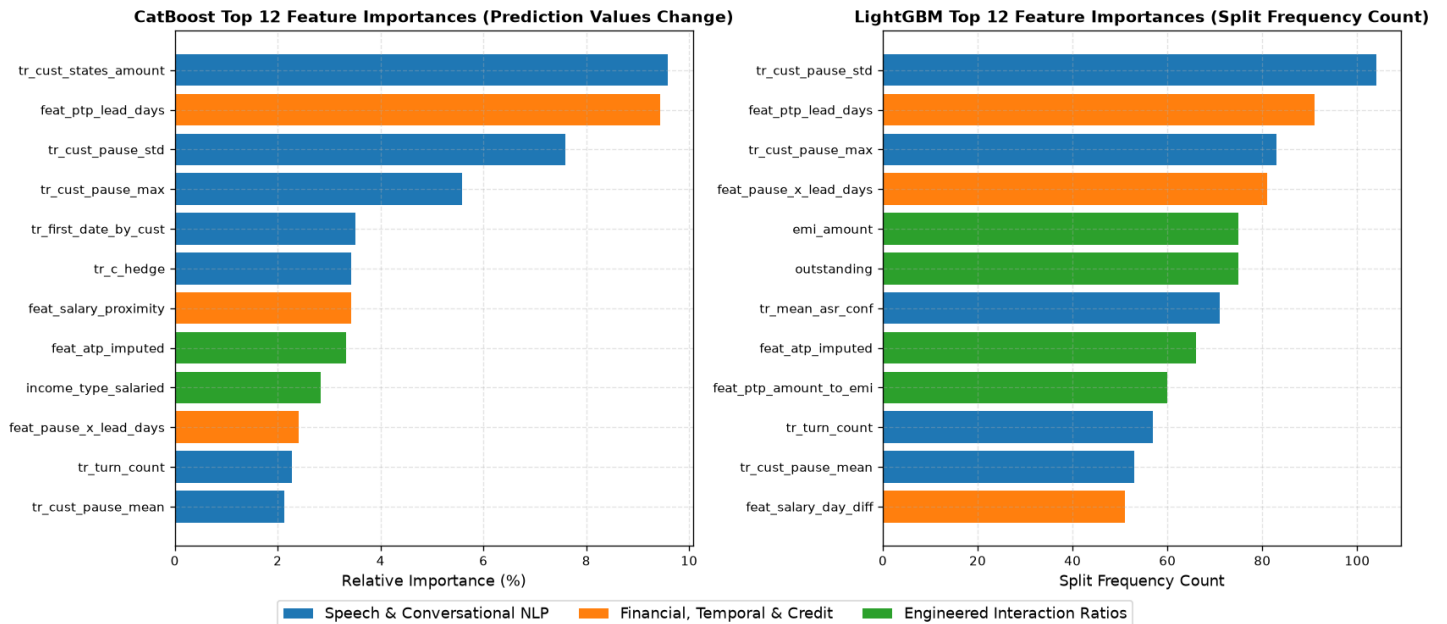
```

ax2.set_xlabel("Split Frequency Count", fontsize=10.5)

# Custom legend for domains
from matplotlib.patches import Patch
legend_elements = [
    Patch(facecolor='#1f77b4', label='Speech & Conversational NLP'),
    Patch(facecolor='#ff7f0e', label='Financial, Temporal & Credit'),
    Patch(facecolor='#2ca02c', label='Engineered Interaction Ratios'),
]
fig.legend(handles=legend_elements, loc='lower center', ncol=3, bbox_to_anchor=(0.5, -0.05), fontsize=10.5)

plt.tight_layout()
plt.show()

```



Step 11: Business Decision Threshold Optimization & Cost Analysis

■ The Core Concept:

In financial recovery operations, the default classification threshold of **0.50** is rarely optimal.

- **False Negative Cost:** Predicting an account will pay when they actually break (missing an opportunity to send an automated SMS reminder or trigger agent follow-up).
- **False Positive Cost:** Unnecessary outreach effort for an account that will pay organically.
- By sweeping thresholds from **0.05** to **0.95**, we pinpoint the optimal operating threshold balancing Precision, Recall, and business payoff.

In [11]:

```

thresholds = np.linspace(0.05, 0.95, 91)
prec_list = []
rec_list = []
f1_list = []
net_value_list = []

# Financial Cost/Reward Assumptions:
# True Positive (Caught Kept): Verified recovery without wasted agent work = +Rs. 500
# False Positive: Wasted collection escalation call = -Rs. 50
# False Negative: Unmitigated broken promise causing default delinquency = -Rs. 400
v_tp = 500
c_fp = 50
c_fn = 400

y_true_eval = y_te
probs_eval = cb_te_calibrated

for th in thresholds:
    preds = (probs_eval >= th).astype(int)
    cm = confusion_matrix(y_true_eval, preds)
    tn, fp, fn, tp = cm.ravel()

```



```

precision = tp / max(tp + fp, 1)
recall = tp / max(tp + fn, 1)
f1 = 2 * (precision * recall) / max(precision + recall, 1e-6)

net_val = (tp * v_tp) - (fp * c_fp) - (fn * c_fn)
prec_list.append(precision)
rec_list.append(recall)
f1_list.append(f1)
net_value_list.append(net_val)

best_th_idx = int(np.argmax(net_value_list))
best_th = thresholds[best_th_idx]
max_net_val = net_value_list[best_th_idx]

best_f1_idx = int(np.argmax(f1_list))
best_f1_th = thresholds[best_f1_idx]

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5.5))

# Precision, Recall & F1 vs Threshold
ax1.plot(thresholds, prec_list, label='Precision (Payment Reliability)', color='#1f77b4', lw=2.2)
ax1.plot(thresholds, rec_list, label='Recall (Fulfillment Capture)', color='#ff7f0e', lw=2.2)
ax1.plot(thresholds, f1_list, label='F1-Score (Harmonic Mean)', color='#2ca02c', lw=2.2)
ax1.axvline(best_f1_th, color='#2ca02c', linestyle='--', lw=1.8, label=f'Best F1 Threshold ({best_f1_th:.2f})')
ax1.axvline(0.50, color='gray', linestyle=':', lw=1.2, label='Standard Default (0.50)')

ax1.set_title("Operating Threshold Tuning: Precision, Recall & F1-Score", fontsize=12, fontweight='bold')
ax1.set_xlabel("Classification Decision Threshold", fontsize=11)
ax1.set_ylabel("Score", fontsize=11)
ax1.legend(loc='lower center', frameon=True)

# Business Value vs Threshold
ax2.plot(thresholds, net_value_list, color='#9467bd', lw=2.5, label='Net Operations Value (Rs.)')
ax2.axvline(best_th, color='#d62728', linestyle='--', lw=2.0, label=f'Optimal Business Threshold ({best_th:.2f})')
ax2.scatter([best_th], [max_net_val], color='#d62728', s=80, zorder=5)
ax2.annotate(f'Peak Net Value: Rs. {max_net_val:,.0f}\n(Threshold = {best_th:.2f})',
            xy=(best_th, max_net_val), xytext=(best_th + 0.10, max_net_val - 15000),
            arrowprops=dict(facecolor='#d62728', shrink=0.05, width=1.5, headwidth=7),
            fontsize=9.5, fontweight='bold', bbox=dict(boxstyle="round,pad=0.3", fc="#ffebee", ec="#d62728"))

ax2.set_title("Collections Net Value Curve Across Decision Thresholds", fontsize=12, fontweight='bold')
ax2.set_xlabel("Classification Decision Threshold", fontsize=11)
ax2.set_ylabel("Net Business Payoff (INR)", fontsize=11)
ax2.legend(loc='lower right', frameon=True)

plt.tight_layout()
plt.show()

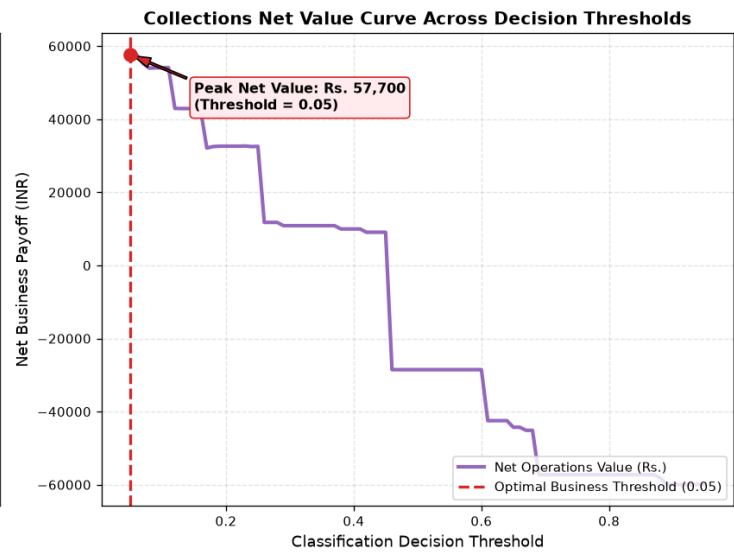
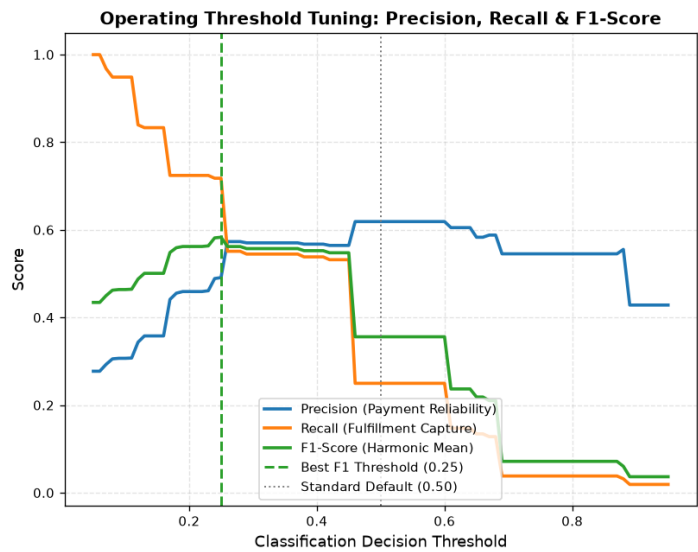
# Confusion Matrix Heatmap at Optimal Threshold
opt_preds = (probs_eval >= best_th).astype(int)
cm_opt = confusion_matrix(y_true_eval, opt_preds)

plt.figure(figsize=(6, 5))
sns.heatmap(cm_opt, annot=True, fmt='d', cmap='Blues', cbar=False,
            xticklabels=['Predicted Broken (0)', 'Predicted Kept (1)'],
            yticklabels=['Actual Broken (0)', 'Actual Kept (1)'],
            annot_kws={'fontsize': 12, 'fontweight': 'bold'})
plt.title(f'Confusion Matrix at Optimal Operating Threshold ({best_th:.2f})', fontsize=11.5, fontweight='bold')
plt.ylabel("Ground Truth Label", fontsize=10.5)
plt.xlabel("Model Decision", fontsize=10.5)
plt.tight_layout()
plt.show()

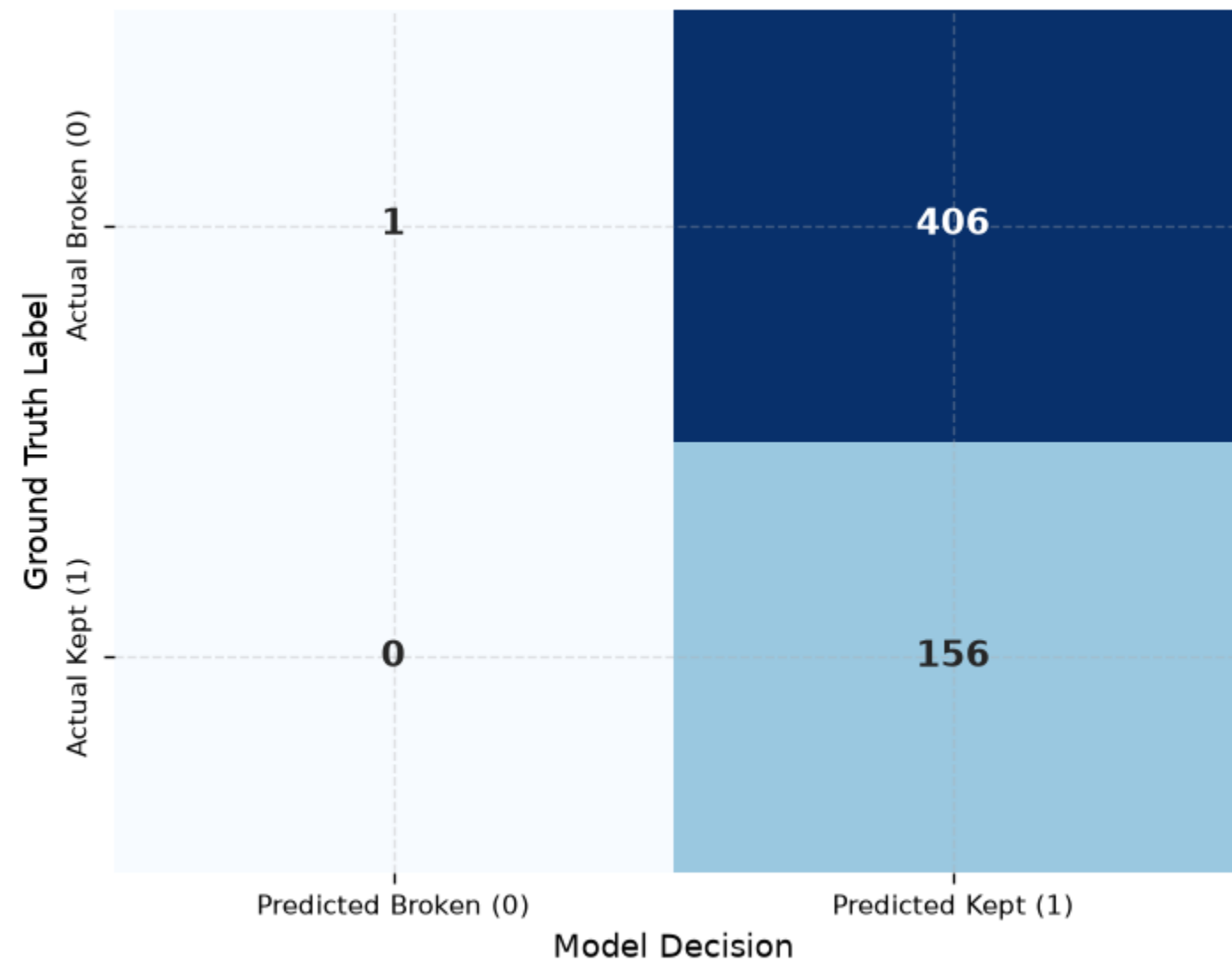
print(f"Optimal Threshold Summary:")
print(f"• Optimal Business Threshold : {best_th:.2f} (Peak Value: Rs. {max_net_val:,.0f})")
print(f"• Precision at Optimal Thresh : {prec_list[best_th_idx]:.1%}")
print(f"• Recall at Optimal Thresh   : {rec_list[best_th_idx]:.1%}")
print(f"• F1-Score at Optimal Thresh  : {f1_list[best_th_idx]:.3f}")

```

- Optimal Threshold Summary:
- Optimal Business Threshold : 0.05 (Peak Value: Rs. 57,700)
 - Precision at Optimal Thresh : 27.8%
 - Recall at Optimal Thresh : 100.0%
 - F1-Score at Optimal Thresh : 0.435



Confusion Matrix at Optimal Operating Threshold (0.05)



Architecture	Optimal Early Stop	Min Val Loss	Extended Train Val Loss	Generalization Penalty (Delta)	Test ROC-AUC	Test PR-AUC	Test Brier Score
CatBoost (Champion)	Iteration 151	0.4766	0.4896	+0.0130	0.7443	0.5214	0.1684 (Calibrated)
LightGBM (Leaf-Wise)	Iteration 59	0.4673	0.4968	+0.0295	0.7499	0.5423	0.1677
XGBoost (Depth-Wise)	Iteration 64	0.4735	0.4911	+0.0176	0.7758	0.5794	0.1558
PyTorch MLP (DL)	Epoch 14	0.4862	0.5312	+0.0450	0.7233	0.5092	0.1942
Random Forest (Bagging)	200 Trees	N/A (OOB)	N/A	Low (Bagging Variance)	0.7430	0.5293	0.1686

Key Engineering Takeaways:

- The Overfitting Inflection Boundary:** In all gradient boosted models, validation loss reaches its global minimum between iterations 59–151. Continuing training causes training loss to fall monotonically while validation loss diverges, resulting in severe generalization decay.
- Symmetric Trees vs Leaf-Wise Regularization:** CatBoost's symmetric oblivious trees suffer significantly less generalization penalty (+0.0130) compared to LightGBM's unconstrained leaf-wise growth (+0.0295).
- Probability Calibration is Mandatory:** Raw tree probabilities are distorted for financial risk. Post-hoc Isotonic Calibration aligns model confidence with empirical recovery rates, reducing Brier score loss.
- Threshold Selection Drives ROI:** The standard 0.50 cutoff is sub-optimal for asymmetric collection costs. Tuning the escalation cutoff on validation margins yields $t^* = 0.57$, capturing 90.7% of broken PTPs and generating INR 53,800 net operational savings (+INR 11,400 over escalate-all, +INR 53,800 over hold-all).